# Xiang Zheng


**Contact Information**

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Google Scholar: 977 citations, h-index 11


**Research Interests**

Reinforcement Learning, Trustworthy AI, Agentic AI, Embodied AI, Trustworthy Recursive Self-Improvement, Agentic Red Teaming, AI for Science

**Education**

**City University of Hong Kong** **2020–2024**
*Ph.D. in Computer Science*, Hong Kong SAR, China
Supervisor: Prof. Cong Wang
Fields: Reinforcement Learning, Trustworthy AI
GPA: 4.00/4.00

**Tsinghua University** **2016–2019**
*M.S. in Control Science and Engineering*, Beijing, China
Supervisor: Prof. Tao Zhang
Fields: Reinforcement Learning, Intelligent Robotics
GPA: 3.55/4.00

**Beihang University** **2012–2016**
*B.S. in Automation, B.S. in Mathematics (Dual Degree)*, Beijing, China
Selected into the Shen Yuan Honors College (top 0.5%, 1 in 200)
GPA: 3.74/4.00

**Research & Work Experience**

**Hong Kong Institute of AI for Science (HKAI-Sci), City University of Hong Kong** **2026–present**
*Research Assistant Professor*, Hong Kong SAR, China
Topic: Self-evolving science agents and trustworthy AI.
– Chief Scientist of SciencePal 2.0

**City University of Hong Kong** **2024.10–2026.03**
*Postdoctoral Fellow*, Hong Kong SAR, China
Supervisor: Prof. Cong Wang
Topic: Reinforcement learning for trustworthy and embodied AI.

**Xi'an Jiaotong University** **2022.09–2023.08**
*Visiting Researcher*, Xi'an, China
Supervisor: Prof. Chao Shen
Topic: Adversarial machine learning.

**Xi'an Jiaotong University** **2019.11–2020.02**

*Visiting Researcher*, Xi'an, China
Supervisor: Prof. Chao Shen and Xingjun Ma
Topic: Adversarial machine learning.

**National Institute of Informatics** **2018.02–2018.05**

*Research Intern*, Tokyo, Japan
Supervisor: Prof. Tetsunari Inamura
Topic: Intelligent robotics.

**The University of New South Wales** **2016.02–2016.06**

*Research Intern*, Sydney, Australia
Supervisor: Prof. Elias Aboutanios
Topic: Spacecraft de-orbiting control.

## Publications

*Conference* (∗ for co-first author, † for corresponding author.)

C-1. Hanxun Huang, Yutao Wu, Qizhou Wang, Silvia Montaña-Niño, Yige Li, **Xiang Zheng**, Elif Buse Doyuran, Phoebe Matich, Xiao Liu, Xingjun Ma, Sarah Erfani, Christopher Leckie, "VEX-Bench: Benchmarking Verification Complexity of LLM-Generated Misinformation," in NeurIPS E&D Track, 2026.

C-2. Yutao Wu, Xiao Liu, Hanxun Huang, Yige Li, **Xiang Zheng**, Yifeng Gao, Cong Wang, Bo Li, Xingjun Ma, Yu-Gang Jiang, "Internal Safety Collapse in Frontier Large Language Models," in NeurIPS, 2026.

C-3. Zhouyuan Ma, Yutao Wu, Hanxun Huang, **Xiang Zheng**, Xiao Liu, Yixin Cao, Zuxuan Wu, Xingjun Ma, Yu-Gang Jiang, "HarmProfile: Characterizing Harmful Distributions in Frontier LLMs," in EMNLP Findings, 2026.

C-4. Xutao Mao, Liangjie Zhao, Tao Liu, **Xiang Zheng**†, Hongying Zan, Cong Wang†, "STARE: Step-wise Temporal Alignment and Red-teaming Engine for Multi-modal Toxicity Attack," in ICML, 2026.

C-5. **Xiang Zheng**, Yutao Wu, Hanxun Huang, Yige Li, Xingjun Ma†, Bo Li, Yu-Gang Jiang, Cong Wang†, "Just Ask: Curious Code Agents Reveal System Prompts in Frontier LLMs," in ICML, 2026.

C-6. Zilong Wang, **Xiang Zheng**†, Xiaosen Wang, Bo Wang, Xingjun Ma†, Yu-Gang Jiang, "GenBreak: Red Teaming Text-to-Image Generators Using Large Language Models," in CVPR, 2026.

C-7. Zonghuan Xu, Jiayu Li, Yunhan Zhao, **Xiang Zheng**†, Xingjun Ma†, Yu-Gang Jiang, "DropVLA: An Action-Level Backdoor Attack on Vision-Language-Action Models," in IROS, 2026.

C-8. Yunhan Zhao, **Xiang Zheng**†, Lin Luo, Yige Li, Xingjun Ma†, Yu-Gang Jiang, "BlueSuffix: Reinforced Blue Teaming for Vision-Language Models Against Jailbreak Attacks," in ICLR, 2025.

C-9. **Xiang Zheng**, Longxiang Wang, Yi Liu, Xingjun Ma, Chao Shen, Cong Wang†, "CALM: Curiosity-Driven Auditing for Large Language Models," in AAAI, 2025.

C-10. **Xiang Zheng**, Xingjun Ma, Chao Shen, Cong Wang†, "Constrained intrinsic motivation for reinforcement learning," in IJCAI, 2024.
Acceptance rate 791/5651 = 14.00%.

C-11. **Xiang Zheng**, Xingjun Ma, Shengjie Wang, Xinyu Wang, Chao Shen, Cong Wang†, "Toward evaluating robustness of reinforcement learning with adversarial policy," in DSN, 2024.
Acceptance rate 42/203 = 20.69%.

C-12. Shengjie Wang, Fengbo Lan, **Xiang Zheng**, Yuxue Cao, Oluwatosin Oseni, Haotian Xu, Tao Zhang, Yang Gao, "A policy optimization method towards optimal-time stability," in CoRL, 2023.

C-13. Shengjie Wang, **Xiang Zheng**, Yuxue Cao, Tao Zhang, "A multi-target trajectory planning of a 6-DoF free-floating space robot via reinforcement learning," in IROS, 2021.

C-14. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "An end-to-end trajectory planning strategy for free-floating space robots," in CCC, 2021.

C-15. Shihao Zhao, Xingjun Ma, **Xiang Zheng**, James Bailey, Jingjing Chen, Yu-Gang Jiang, "Clean-label backdoor attacks on video recognition models," in CVPR, 2020.
Acceptance rate 1470/6656 = 22.09%.

C-16. **Xiang Zheng**, Ziwei Wang, Tao Zhang, "Robust finite-time attitude tracking control for nonlinear quadrotor with uncertainties and delays," in ROBIO, 2017.

*Journal*

J-1. Yunhan Zhao, **Xiang Zheng**, Yige Li, Xingjun Ma†, "Defense-to-attack: Bypassing weak defenses enables stronger jailbreaks in Vision-Language Models," Pattern Recognition, 2026, art. no. 113805.
Impact Factor 7.6; JCR Rank Q1.

J-2. Yi Liu, **Xiang Zheng**, Chengjun Cai, Xingliang Yuan, Cong Wang†, "Adapting Large Language Models for Encrypted Traffic Analysis Services: An Efficient Realization with Mixture of LoRA Experts," IEEE Transactions on Services Computing, vol. 19, no. 2, pp. 906–919, 2026.
Impact Factor 5.5; JCR Rank Q1.

J-3. **Xiang Zheng**, Xingjun Ma, Wei-Bin Lee, Cong Wang†, "OpenRedRL: A Light-Weight Benchmark for Reinforcement Learning-Based Red Teaming," Frontiers of Computer Science, 2026.
Impact Factor 4.6; JCR Rank Q2.

J-4. Xingjun Ma, Yifeng Gao, Yixu Wang, . . . , **Xiang Zheng**, . . . , Yu-Gang Jiang (48 authors), "Safety at Scale: A Comprehensive Survey of Large Model and Agent Safety," Foundations and Trends in Privacy and Security, vol. 8, no. 3-4, pp. 254–469, 2026.

J-5. Yuxue Cao, Shengjie Wang, **Xiang Zheng**, Wenke Ma, Xinru Xie, Lei Liu, "Reinforcement learning with prior policy guidance for motion planning of dual-arm free-floating space robot," Aerospace Science and Technology (AST),

vol. 136, pp. 108098, 2023.
Impact Factor 5.6; JCR Rank Q1 (3/34).

J-6. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "A learning system for motion planning of free-float dual-arm space manipulator towards non-cooperative object," Aerospace Science and Technology (AST), vol. 131, pp. 107980, 2022.
Impact Factor 5.6; JCR Rank Q1 (3/34).

J-7. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "Collision-free trajectory planning for a 6-DoF free-floating space robot via hierarchical decoupling optimization," IEEE Robotics and Automation Letters (RA-L), vol. 7, no. 2, pp. 4953–4960, 2022.
Impact Factor 5.2; JCR Rank Q2 (10/30).

PREPRINTS

1. Xutao Mao, Liangjie Zhao, **Xiang Zheng**†, Cong Wang†, "Practice Makes Unsafe: Skill Misevolution in Self-Improving LLM Agents," arXiv preprint arXiv:2608.12851.

2. Xutao Mao, **Xiang Zheng**†, Cong Wang†, "Agent Hacks Agent: Autoresearch for Production-Agent Red-Teaming," arXiv preprint arXiv:2607.11698.

3. Xutao Mao, Liangjie Zhao, Leyao Wang, Rui Qian, Qiang Huang, Wentao Wang, Bo Han, **Xiang Zheng**†, Cong Wang†, "Agents Don't Just Agree, They Remember: Benchmarking Persistent Sycophancy in Stateful Personal Agents," arXiv preprint arXiv:2607.10526.

4. Yige Li, Jun Sun, Wei Zhao, Zhe Li, Yutao Wu, Hanxun Huang, **Xiang Zheng**, Xingjun Ma, "When Medical Safety Alignment Fails: A Benchmark for Evaluating LLMs on High-Risk Medical Queries," arXiv preprint arXiv:2606.28332.

5. Xiao Li*, **Xiang Zheng***, Yifeng Gao, ..., Xingjun Ma†, Yu-Gang Jiang† (38 authors), "Safety in Embodied AI: A Survey of Risks, Attacks, and Defenses," arXiv preprint arXiv:2605.02900.

6. Longxiang Wang, **Xiang Zheng**†, Xuhao Zhang, Yao Zhang, Ye Wu, Cong Wang†, "OptiLeak: Efficient Prompt Reconstruction via Reinforcement Learning in Multi-tenant LLM Services," arXiv preprint arXiv:2602.20595.

7. Zixing Chen, Yifeng Gao, Li Wang, Yunhan Zhao, Yi Liu, Jiayu Li, **Xiang Zheng**, Zuxuan Wu, Cong Wang, Xingjun Ma, Yu-Gang Jiang, "HazardArena: Evaluating Semantic Safety in Vision-Language-Action Models," arXiv preprint arXiv:2604.12447.

8. Zonghuan Xu, **Xiang Zheng**, Yutao Wu, Xingjun Ma, "When Direct Prediction Fails: Evidence from LLM-Based Misinformation Risk Evaluation," arXiv preprint arXiv:2604.06820.

9. Ming Wen, Yuxuan Liu, Kun Yang, Yunhao Feng, Zhuoer Xu, Yuhao Sun, Shiwen Cui, **Xiang Zheng**, Yi Liu, Xingjun Ma, Yu-Gang Jiang, "Constitutional On-Policy Safe Distillation," arXiv preprint arXiv:2606.03089.

10. Jiayu Li, Yunhan Zhao, **Xiang Zheng**, Zonghuan Xu, Yige Li, Xingjun

Ma, Yu-Gang Jiang, "AttackVLA: Benchmarking Adversarial and Backdoor Attacks on Vision-Language-Action Models," arXiv preprint arXiv:2511.12149.

11. Yutao Wu, Xiao Liu, Yinghui Li, Yifeng Gao, Yifan Ding, Jiale Ding, **Xiang Zheng**, Xingjun Ma, "ADMIT: Few-shot Knowledge Poisoning Attacks on RAG-based Fact Checking," arXiv preprint arXiv:2510.13842.

12. Jiale Ding, **Xiang Zheng**†, Yutao Wu, Cong Wang, Wei-Bin Lee, Ling Pan, Xingjun Ma†, Yu-Gang Jiang, "RedTopic: Toward Topic-Diverse Red Teaming of Large Language Models," arXiv preprint arXiv:2507.00026.

13. Yuxue Cao, Wenhao Zhao, Shengjie Wang, **Xiang Zheng**, Wenke Ma, Ziwei Wang, Tao Zhang, "A Learning-based Adaptive Compliance Method for Symmetric Bi-manual Manipulation," arXiv preprint arXiv:2303.15262.

## Grants & Projects

*Independent Grants as PI or Co-PI*

**Theme-based Research Scheme (TRS), Research Grants Council of Hong Kong** **2027/28**

Co-PI; In preparation.

*Co-written Grants*

**Foxconn-CityUHK Joint Research Centre** **2027**

2027 Red Teaming Research Proposal (HKD 500K requested)
Proposal Co-writer; Submitted.

**Theme-based Research Scheme (TRS), Research Grants Council of Hong Kong** **2026/27**

SynapseCity: AI-Driven Urban Emergency Response and Rescue HKD 55.691M
Proposal Writer and Multi-Agent Prototype Contributor; Awarded.

**Foxconn-CityUHK Joint Research Centre** **2026**

Red Teaming Technology Research for Intelligent Agents HKD 500K
Proposal Co-writer; In progress.

**Foxconn-CityUHK Joint Research Centre** **2025**

Intrinsically Motivated RFT for Red Teaming Black-Box LLMs HKD 500K
Proposal Co-writer.

*Participated Research Projects*

**CRF, No. 8730083** **2023–present**

Enabling Metadata-Private and Accountable Networks at Scale HKD 7.52M

**NSFC, No. 9054034** **2022–present**

Towards Secure and Privacy-enhanced Machine Learning as a Service CNY 1M

**RFS, No. 9062004** **2022–present**
Building Privacy-Assured and Scalable Encrypted Databases With Secure Enclave HKD 5.16M

**CAST** **2017–2019**
Learning-Based Control Technology CNY 1.8M

## Invited Talks

**Medical Large Model Agent Forum, CHIP 2026** **2026.10.31**
Dalian
*SciencePal 2.0 Research Agent*

**Auto-Research Forum, CNCC 2026** **2026.10**
Chengdu
*SciencePal 2.0*

**Southern University of Science and Technology (Prof. Jianyu Niu's Group)** **2026.09.03**
Online
*Trustworthy Recursive Self-Improvement*

**Huawei Hong Kong Research Institute** **2026.08.27**
Hong Kong
*Towards Trustworthy Agents: From Red Teaming to Trustworthy Recursive Self-Improving*

**CityUHK Joint Summer School on AI for Science** **2026**
Dongguan
*Hands-on with Claude Code: Frontier Code Agents for Research and Browser/Computer Use*

**Tencent Cloud AI Industry Applications Summit** **2026**
Beijing
*SciencePal 2.0: Your Self-Evolving Science Agent Team*

**HKAI-Sci, CityUHK** **2026**
Hong Kong
*Build Your Own Self-Evolving Science Agent with SciencePal 2.0*

**ByteDance** **2026**
Shenzhen
*Agentic Red Teaming and Blue Teaming*

**College of Computing at CityUHK (with Prof. Cong Wang)** **2026**
Hong Kong
*Is OpenClaw All You Need?*

**Lancaster University (Prof. Ziwei Wang's Group)** **2026**
Online

*From Robot Learning to Self-Evolving OpenClaw: Building and Securing AI Agent Systems*

**HKAI-Sci, CityUHK** **2026**
Hong Kong
*Harness Engineering*

**Institute of Trustworthy Embodied AI at Fudan (Prof. Xingjun Ma's Group)** **2026**
Online
*Reinforcement Learning for Agent Safety Evaluation*

**CCF YOCSEF** **2025**
Xi'an
*Towards Robust Embodied AI: Skill Discovery and Red Teaming*

**Southern University of Science and Technology (Prof. Jianyu Niu's Group)** **2025**
Shenzhen
*Reinforcement Learning-Based Adversarial Safety Evaluation and Defense Enhancement for Large Language Models*

**CCF YOCSEF** **2025**
Xi'an
*Reinforcement Fine-Tuning for Large Model Red-/Blue-Teaming*

**School of Computer Science at Northwestern Polytechnical University (Prof. Helei Cui's Group)** **2024**
Xi'an
*Curiosity-Driven Auditing for LLMs*

**School of Computer Science at Northwestern Polytechnical University (Prof. Helei Cui's Group)** **2024**
Xi'an
*Efficient Intrinsically Motivated Adversarial Policy Learning*

**Griffith University (Prof. Leo Yu Zhang's Group)** **2024**
Online
*Intrinsically Motivated Adversarial Policy*

**School of Computing and Information Systems at the University of Melbourne (Prof. Xingliang Yuan's and Prof. Feng Liu's Groups)** **2024**
Melbourne
*Towards Efficient Evasion Attacks Against RL*

## Awards & Honors

**ICML Gold Reviewer Award** **2026**
Top 25%

**IJCAI Travel Grant** **2024**
IJCAI Organization and AIJ Division
USD 500

**International Conference Grant** **2024**
City University of Hong Kong
HKD 10,000

**DSN Student Travel Grant** **2024**
DSN Student Travel Awards Committee
USD 1,500

**Research Activities Fund** **2022–2023**
City University of Hong Kong
HKD 96,000

**Institutional Research Tuition Grant** **2020–2024**
City University of Hong Kong
HKD 168,384

**CityUHK Presidential PhD Scholarship** **2020–2024**
2/73
HKD 1,353,120

**NII MOU Research Activities Grant** **2018**
1/106
National Institute of Informatics, Japan
JPY 68,400

**CSC Scholarship** **2016**
China Scholarship Council
AUD 6,400

**Stars of Advanced Engineering** **2015**
Shen Yuan Honors College, Beihang University
Top-2 of 50 honors students
Highest honor at the Honors College

## Teaching

CS5293, Topics on Information Security
CS4394, Information Security and Management
CS4293, Topics in Cybersecurity
CS5294, Information Security Technology Management
CS6290, Privacy-enhancing Technologies
CS2310/CS2311, Computer Programming

## Mentored Students

**PhD**
Longxiang Wang (City University of Hong Kong), RL, AI for Privacy, Agent Safety
Pinsheng Xu (City University of Hong Kong), Multi-Agent System Safety

Xutao Mao (City University of Hong Kong), RL, VLM Safety, Agent Safety [ICML'26]
Ruofan Wang (Fudan University), RL, Red Teaming for VLM
Zilong Wang (Fudan University), RL, Red Teaming for Text-to-Image Model [CVPR'26]

**Master**
Chenghao Yao (City University of Hong Kong), RL, Unlearning
Shuxuan Lye (City University of Hong Kong), GUI Agent Safety
Xiao Hu (City University of Hong Kong), RL, Agent-as-Judge
Xiao Li (Fudan University), Multi-Agent RL Safety
Yunhan Zhao (Fudan University), RL, Blue Teaming for VLM [ICLR'25; PR'26; UIUC PhD '26]
Zixing Chen (Fudan University), RL, VLA Safety
Jiayu Li (Fudan University), VLA Safety
Shengjie Wang (Tsinghua University), RL, Safe RL, Robot Learning, Space Robot Control [IROS'21; RA-L'22; AST'22/'23; CoRL'23; DSN'24; Tsinghua PhD '22]
Yuxue Cao (Institute of Automation), RL, Robot Learning, Space Robot Control [IROS'21; RA-L'22; AST'22/'23; CoRL'23; Siemens]
Siyuan Zhang (Northwestern Polytechnical University), Agent for Crowdsensing

**Undergraduate**
Jiale Ding (Fudan University), RL, LLM Safety [HKUST PhD '26 (HK PhD Fellowship)]
Zonghuan Xu (Fudan University), VLA Safety, Misinformation [IROS'26]

**Research Assistant**
Cheng Gui (City University of Hong Kong), Recursive Self-Improvement Safety

## Professional Activity

**Program Committee Member**
NeurIPS 2026, ICML 2026, ICLR 2025/2026/2027, CVPR 2026, ECCV 2026, EMNLP 2026, AAAI 2025/2026/2027, MM 2025/2026, ICRA 2026

**Organization**
Special Session Organizer: PRCV 2026

**Journal Reviewer**
CSUR, TPAMI, TDSC, TSC, TC, TMLR, IoT-J

**External Conference Reviewer**
ACL 2026, NeurIPS 2025, ICNP 2025, ESORICS 2022, AsiaCCS 2022, RAID 2021